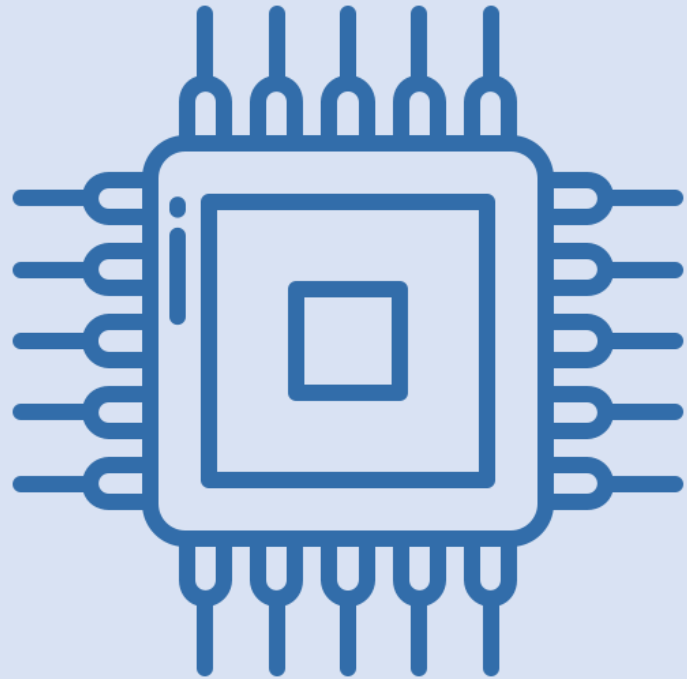


AIN SHAMS UNIVERSITY

Basics of Mechatronics – MCT231



Major Task Team #2



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24P0098**

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1 Work Breakdown Structure (WBS)

1. Project Management & Preliminary Design

- **1.1 Team Organization:** Assign roles (e.g., Mechanical Lead, Electronics Lead, Firmware Lead).
- **1.2 Planning:** Create a **Gantt Chart** to manage the project timeline.
- **1.3 Product Design Specification (PDS):** Define the technical and functional requirements for the toy.
- **1.4 User Research:** Conduct research for the 2 – 10 age group to determine emotional impact and user needs.

2. Design Thinking & Ideation

- **2.1 Define the Challenge:** Formulate a well-defined design challenge based on research.
- **2.2 Conceptualization:** Generate ideas via mind maps and mood boards.
- **2.3 Sketching:** Create initial hand-drawn sketches of the toy concept.
- **2.4 Feedback Loop:** Conduct iterative design cycles based on lab feedback.

3. Mechanical Design & Analysis

- **3.1 Mechanism Development:**
 - **3.1.1 Projectile-Based Shooting Mechanism:** Design the launcher
 - **3.1.2 Vertical Lifting Mechanism:** Design the elevator or claw to move marbles/balls.
 - **3.1.3 Sorting Mechanism:** Design a system to sort by color
- **3.2 CAD Modeling:** Create 3D models and assembly drawings in **Autodesk Inventor**.
- **3.3 Mechanical Calculations:** Perform calculations for projectile motion and actuator sizing.
- **3.4 Safety Integration:** Ensure all moving parts are safe and the unit fits within the 100 cm x 100 cm limit.

4. Mechatronics & Control Systems

- **4.1 Hardware Selection:** Buy necessary hardware
- **4.2 Electronics Design:**
 - **4.2.1 Schematic Design:** Create circuit diagrams in **Proteus**.
 - **4.2.2 Custom PCB:** Design and manufacture at least one custom PCB.
 - **4.2.3 Sensor System:** Integrate sensors for sorting and operation.
- **4.3 User Interfaces:**
 - **4.3.1 Primary Interface:** Program the **LCD/TFT Display** connected to Arduino.
 - **4.3.2 Secondary Interface:** Develop a **Bluetooth-connected application**
- **4.4 Firmware Development:** Write Arduino code for a closed-loop system including start/stop functions.

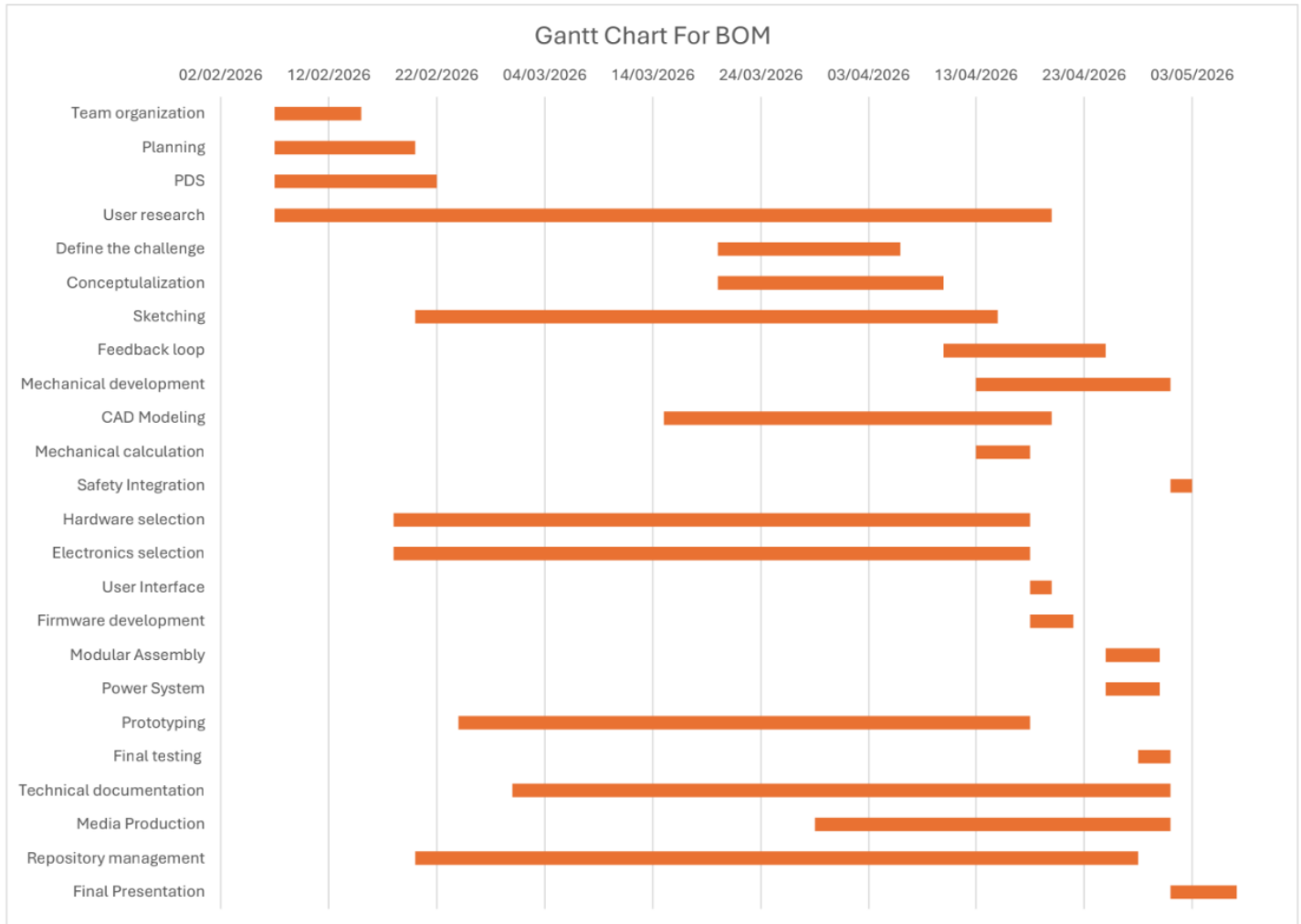
5. Fabrication & Integration

- **5.1 Modular Assembly:** Build the toy in modules with dedicated wiring plugs for easy assembly/disassembly.
- **5.2 Power System:** Integrate a single power source and manage current consumption.
- **5.3 Prototyping:** Build the physical model using appropriate safe materials.
- **5.4 Final Testing:** Conduct full system testing to ensure a "closed loop" marble pathway.

6. Deliverables & Documentation

- **6.1 Technical Documentation:** Finalize the Technical Report and Bill of Materials.
- **6.2 Media Production:** Record a project video (with optional voiceover) and design an **A0 Poster**.
- **6.3 Repository Management:** Organize the **Git repository** with mechanical, electrical, and firmware folders.
- **6.4 Final Presentation:** Prepare and deliver the group presentation.

2 Gantt Chart



3 Product Design Specification (PDS)

1. Performance & Functional Requirements

- **Primary Function:** An interactive basketball-themed toy where a ball is lifted, sorted, and shot into a hoop.
- **Sorting Mechanism:** A sensor-based system designed to differentiate balls specifically by **color**.
- **Lifting Mechanism:** A **continuously rotating spiral screw** (Archimedes' screw) to transport balls vertically.
- **Shooting Mechanism:** A projectile system calibrated to launch the ball into a basketball hoop.
- **Closed-Loop System:** A single-entry point for marbles/balls that must remain within the internal track throughout the cycle.
- **Interactive Features:** Must include at least one embedded interactive feature (e.g., buttons or sensors) and a dedicated Start/Stop button.

2. Electronics & Control System

- **Microcontrollers:** Two microcontrollers (e.g., Arduino Uno) using a standard communication protocol.
- **User Interface 1:** An onboard **LCD/TFT Touch Display** for direct interaction (No 7-segments).
- **User Interface 2:** A remote application built in **Processing IO** communicating via **Bluetooth**.
- **Custom Electronics:** At least one custom-manufactured **PCB** must be integrated into the system.
- **Power:** Powered by a **single unified power source**.

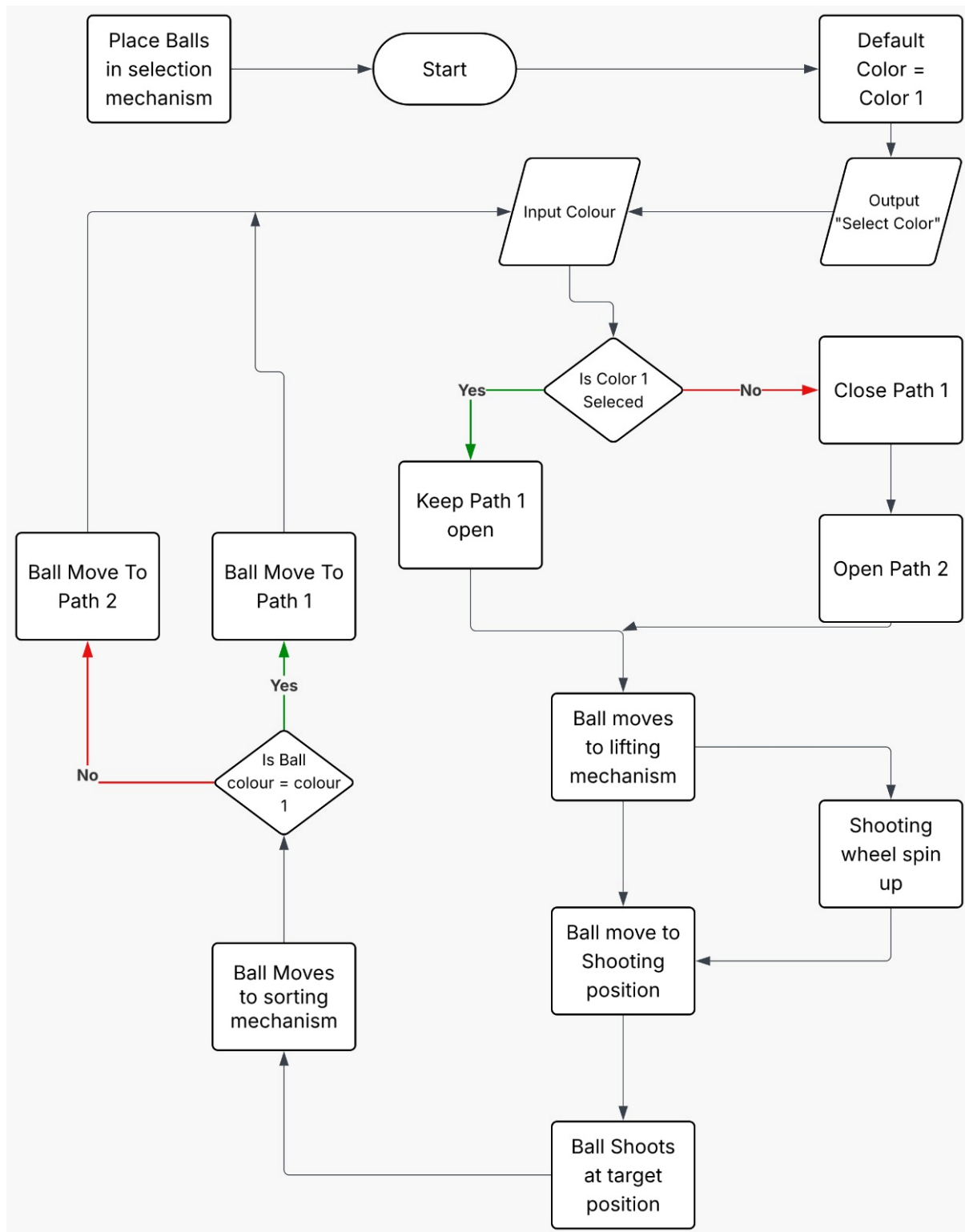
3. Physical & Material Specifications

- **Manufacturing:** Primary structural and mechanical components will be **3D printed**.
- **Dimensions:** Total assembly must not exceed **100 cm x 100 cm**.
- **Modularity:** The design must be divided into modules with individual wiring plugs for easy assembly and disassembly.
- **Safety:** All wiring must be hidden; the 3D-printed surfaces must be smooth and safe for the target age group.

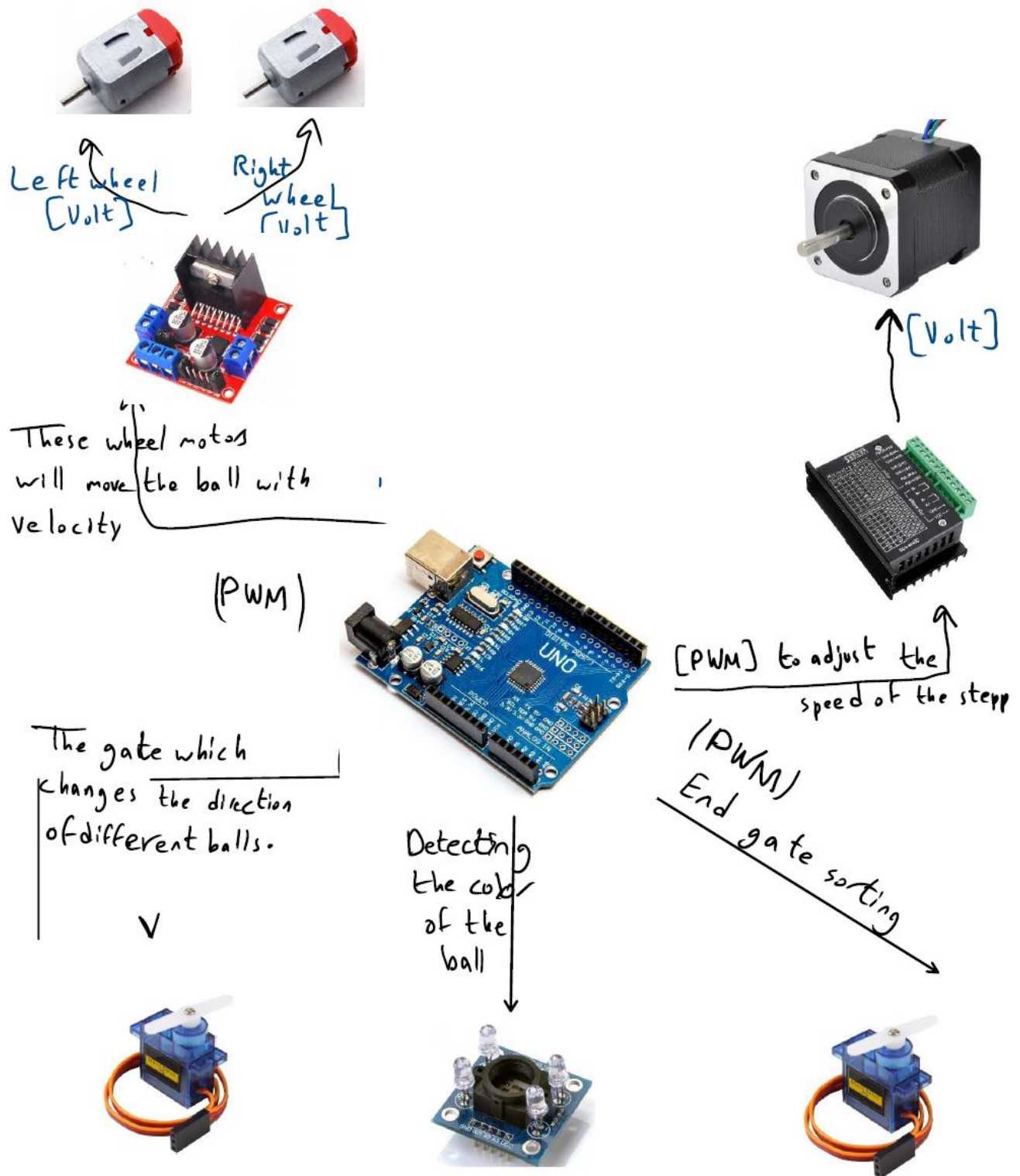
4. Target Audience

User Profile: Designed for children aged 8 years old (\$\pm\$ 6), focusing on engagement levels suitable for older children while maintaining safety for younger ones.

4 System Flowchart

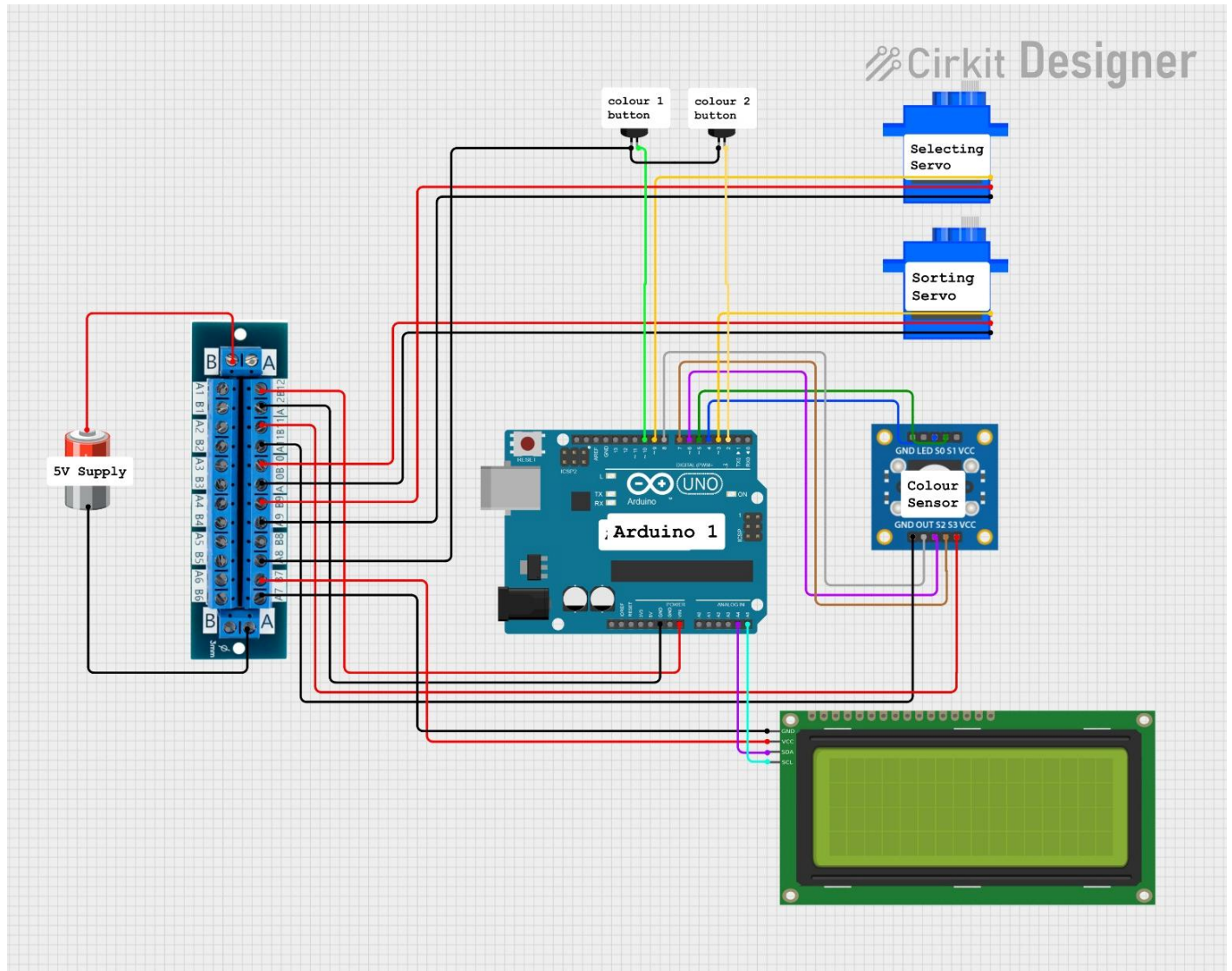


5 Block Diagram

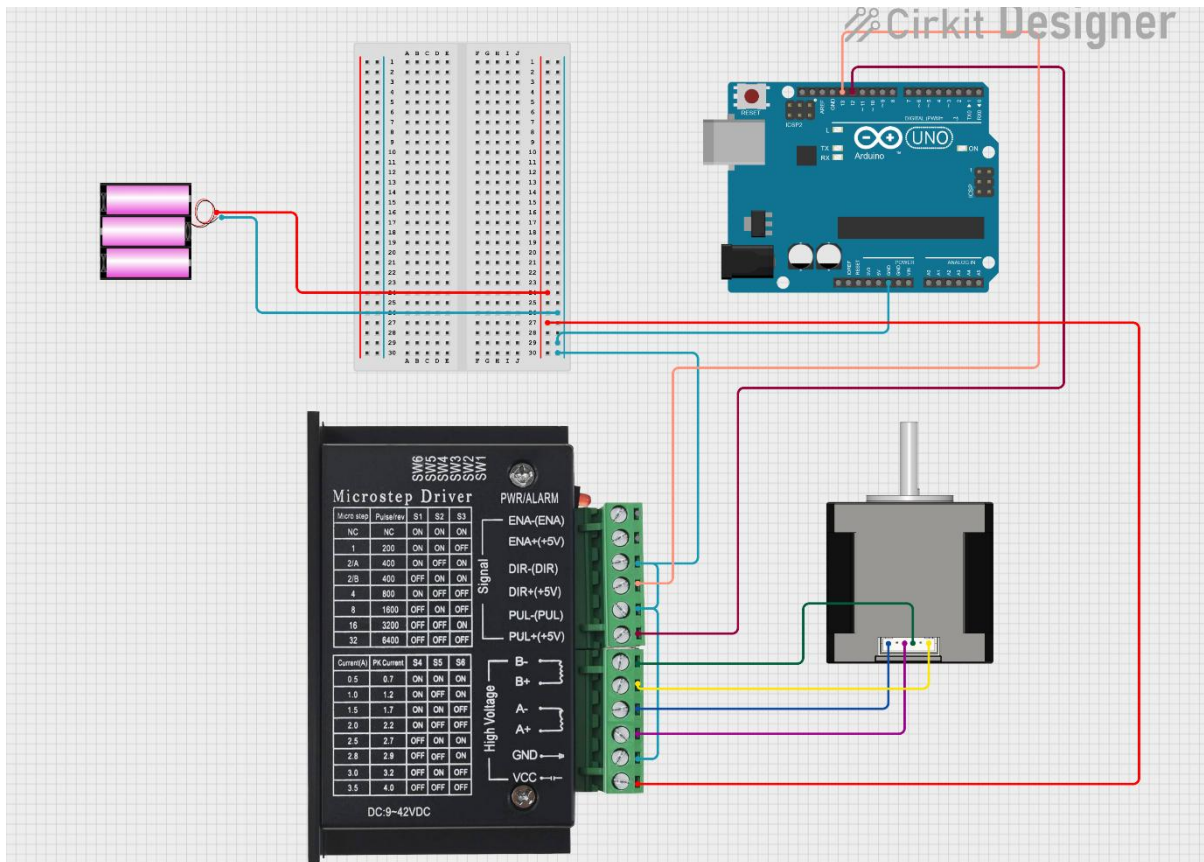


6 Wiring Diagrams

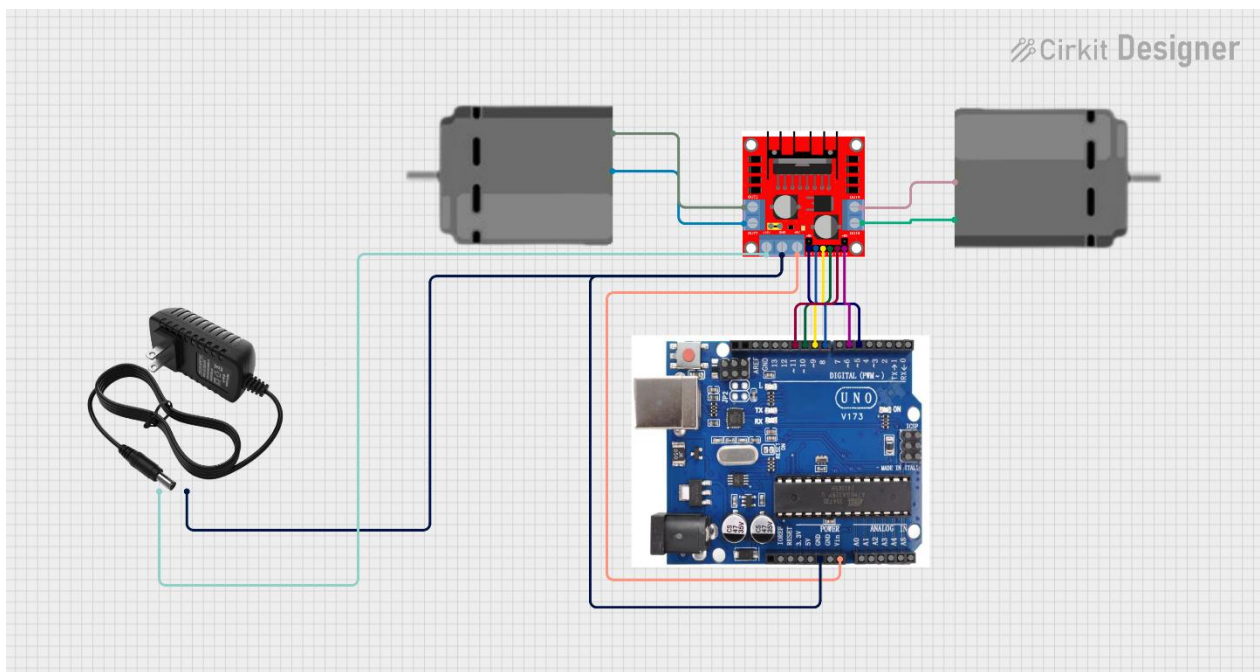
6.1 Sorting:



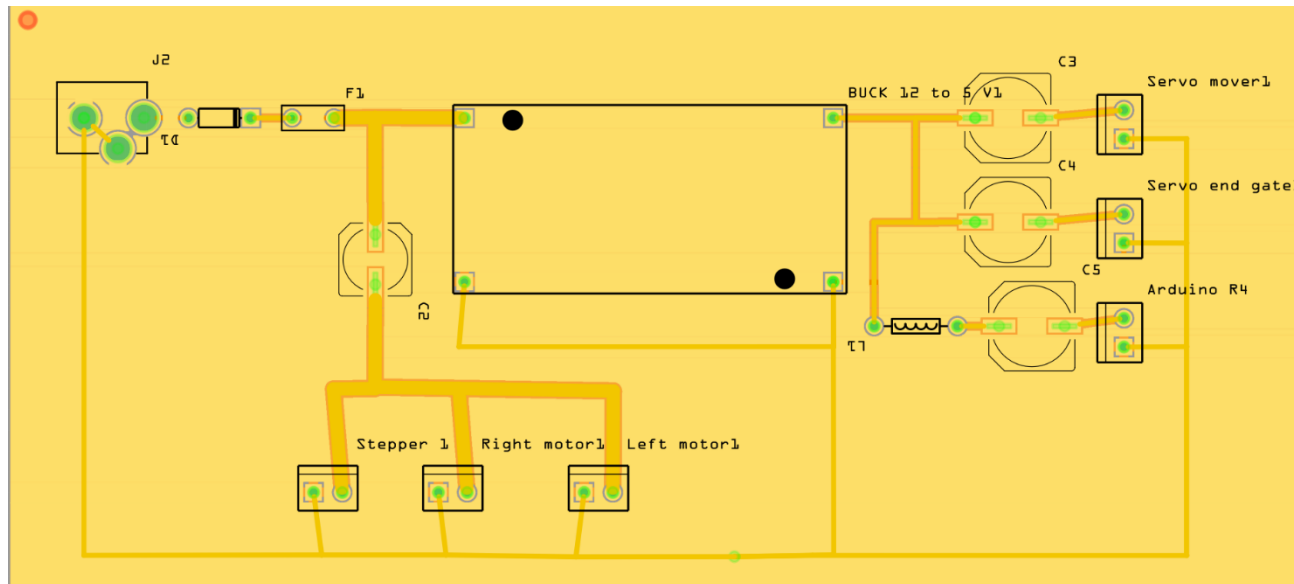
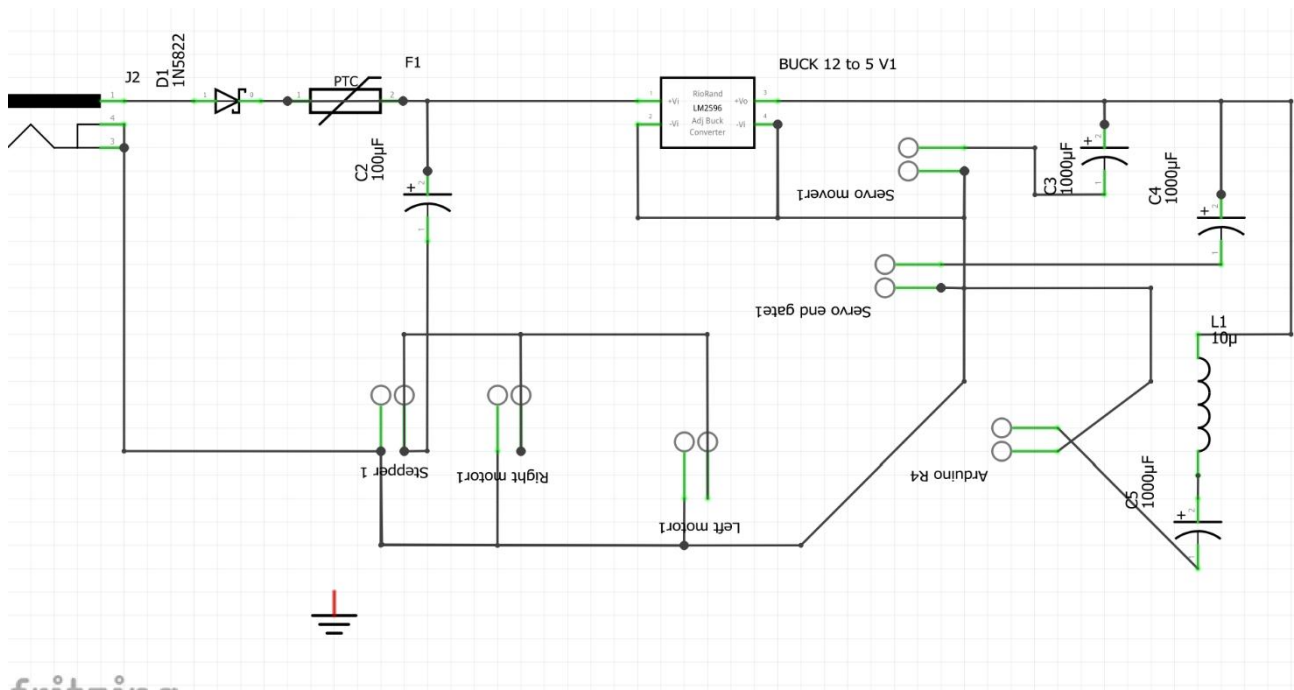
6.2 Lifting:



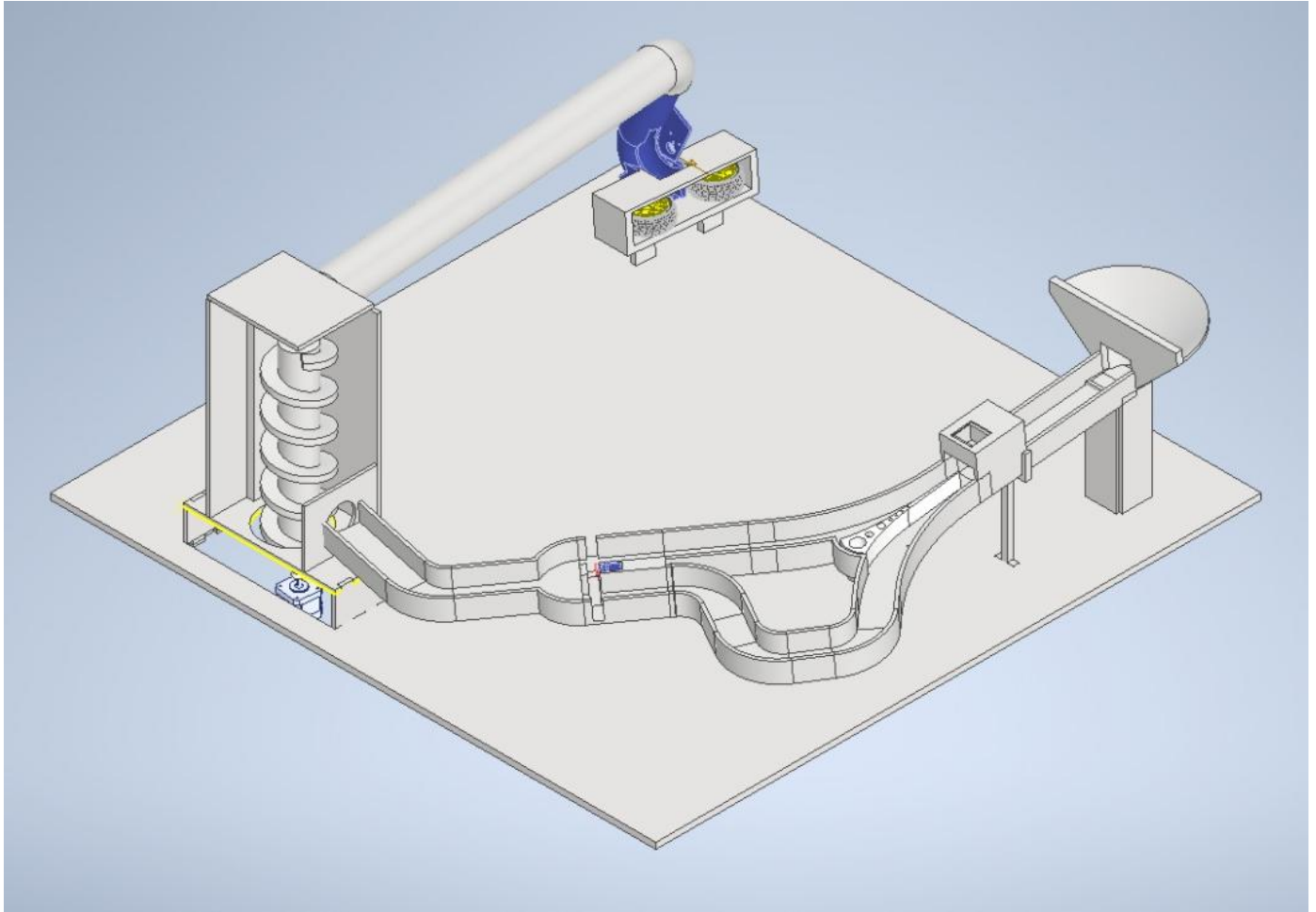
6.3 Shooting:



7 PCB Schematic



8 Fully integrated CAD Design



9 Actuator Sizing

9.1 Sorting:

Sorting Mechanism:

moving Part:

1- Inertia:

$$J_L = J_{arm} + J_{mover} + J_{shaft}$$

$$J_{shaft} = 2.95 \times 10^{-7} \text{ Kg.m}^2$$

$$J_{mover} = 0.000100451 \text{ Kg.m}^2$$

$$J_{arm} = 3.44 \times 10^{-9} \text{ Kg.m}^2$$

$$J_{tot} = \frac{11}{10} J_L = 1.10826584 \times 10^{-4} \approx 1.11 \times 10^{-4} \text{ Kg.m}^2$$

$$\rightarrow T_m = J_{eq} \ddot{\theta}_m + T_R$$

$$\frac{60^\circ}{0.1} = 600^\circ/\text{s}$$

$$\ddot{\theta}_m = 523.2 \text{ rad/s}^2$$

$$\dot{\theta} = \frac{10}{3} \pi \text{ rad/s}$$

$$\ddot{\theta} = \frac{10}{3 \times 0.02} \pi = 523.2 \text{ rad/s}^2$$

$$T_m = 523.2 \times 1.11 \times 10^{-4} + T_R$$

$$T_m \approx 0.0591 + T_R \text{ N.m}$$

$$T_m \approx 0.0591 \text{ N.m}$$

$$T_{m_{ref}} = 0.591 \text{ Kg.cm}$$

* $T_R = 0$, no resisting force $\rightarrow$ no force applied $\rightarrow$ minimal friction

• Motor Used:

$\rightarrow$ Micro servo Motor 9g SG-90:

$$\rightarrow T_s = 1.3 \text{ Kg.cm}$$

$$n = \frac{1.3}{0.591} \approx 2.42 \text{ safety factor}$$

$\rightarrow$ washer is used



* Readings obtained using Autodesk Inventor Iproperties

9.2 Lifting:

Lifting Mechanism:

1 - Inertia:

$$J_L = J_{\text{coupler}} + J_{\text{screw}}$$

$$J_{\text{screw}} = 0.000551 \text{ Kg.m}^2$$

$$J_{\text{coupler}} = 8.52 \times 10^{-7} \text{ Kg.m}^2$$

$$J_L \approx 5.513 \times 10^{-4} \text{ Kg.m}^2$$

$$J_{\text{motor}} = 5.7 \times 10^{-6} \text{ Kg.m}^2$$

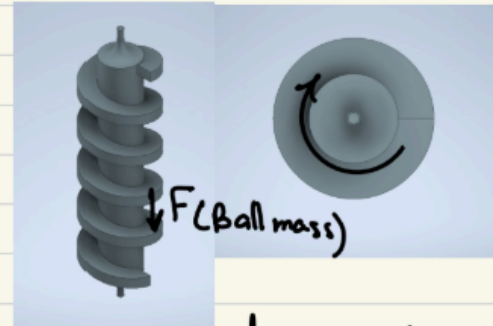
$$J_{\text{tot}} = 5.567 \times 10^{-4} \text{ Kg.m}^2$$

$$T_m > \ddot{\theta} J_{\text{tot}} + T_R$$

$$T_m > 314.1 \times 5.567 \times 10^{-4}$$

$$T_m > 0.175 \text{ N.m}$$

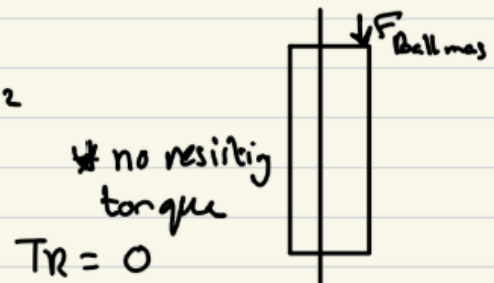
Moving Part:



$$t_{\text{acc}} = 0.1 \text{ s}$$

$$\dot{\theta} = 31.41 \text{ rad/s}$$

$$\ddot{\theta} = 314.1 \text{ rad/s}^2$$



→ Chosen Model : NEMA 17 Stepper Motor
 $T_{\text{available}} = 0.48 \text{ N.m}$

$$n = \frac{0.48}{0.175} = 2.743 \text{ safety Factor}$$

Valid motor, maximum operating speed
 will be around $500 \pm 50 \text{ rpm}$,

* constant Torque

9.3 Shooting:

Shooting Mechanism:

1- Inertia:

$$J_{tot} = J_m + J_w + J_B$$

$$J_m \approx 1.27 \times 10^{-7} \text{ Kg} \cdot \text{m}^2$$

$$J_w \approx 3.69 \times 10^{-5} \text{ Kg} \cdot \text{m}^2$$

$$J_B \approx 7.2 \times 10^{-7} \text{ Kg} \cdot \text{m}^2$$

$$J_{tot} = 3.7247 \times 10^{-5} \text{ Kg} \cdot \text{m}^2$$

$$J_{tot} \approx 3.78 \times 10^{-5} \text{ Kg} \cdot \text{m}^2$$

$$t_{spin\ up} = 3 \text{ sec}$$

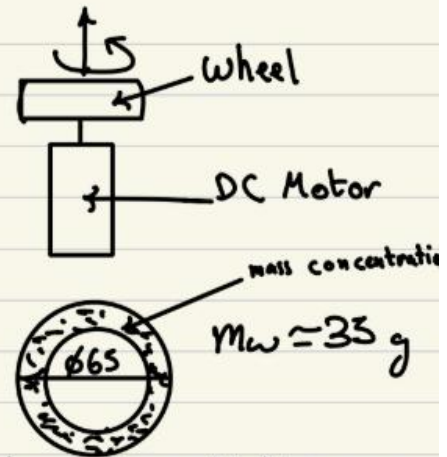
$$\ddot{\theta} = \frac{182.5}{3} \approx 60.8$$

$$T_{m\ req} = J \ddot{\theta} + T_R$$

$$T_{m\ req} = 60.8 \times 3.78 \times 10^{-5} +$$

$$T_{m\ req} = 2.29824 \times 10^{-3}$$

$$T_m \approx 2.3 \times 10^{-3} \text{ N} \cdot \text{m}$$



Wheel PLAN VIEW

$$J_w = m \cdot r^2$$

$$J_w = 0.035 \cdot \left(\frac{65 \times 10^{-3}}{2}\right)^2$$

$$J_w \approx 3.69 \times 10^{-5} \text{ Kg} \cdot \text{m}^2$$

final motor speed:

$$n = 1,743 \text{ rpm}$$

$$\dot{\theta} = 182.5 \text{ rad/s}$$

$$\ddot{\theta} = 60.8 \text{ rad/s}^2$$

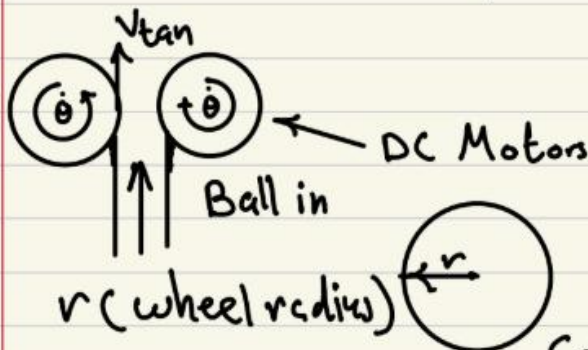
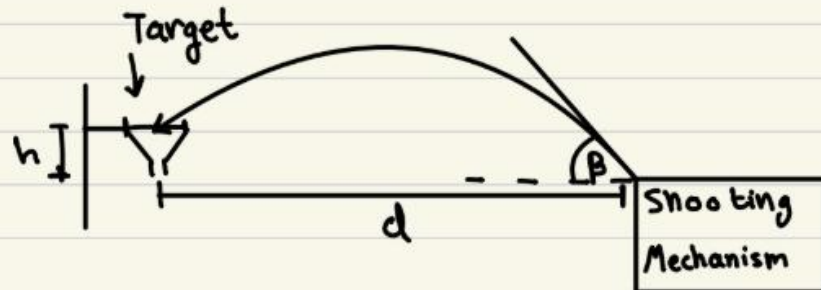
Will use a FA-130RA-18100 Motor, combined with an H-Bridge (2298N) and PWM to achieve required torque and angular speed

Projectile Calculations

$$\beta = 30^\circ$$

$$d = 0.8 \text{ m}$$

$$h = 0.2 \text{ m}$$



DC Motors Wheel details:

$$r = 32.5 \text{ mm} = 32.5 \times 10^{-3} \text{ m}$$

$$m_w = 35 \text{ g} = 3.5 \times 10^{-2} \text{ Kg}$$

Calculation:

$$v_{\tan} = \dot{\theta} r$$

$$\eta \approx 0.7$$

$$v_{0 \text{ req}} = \frac{0.8}{\cos 30} \cdot \sqrt{\frac{9.81}{2(0.8 \tan 30 - 0.2)}}$$

$$v_0 \approx 4.15 \text{ m/s}$$

$$v_{\tan} = \frac{4.5}{0.7} \quad \eta \approx 0.7$$

$$v_{\tan} = 5.93 \text{ m/s}$$

$$\dot{\theta} = \frac{5.93}{0.0325} = 182.5 \text{ rad/s}$$

Conclusion:

$$d = 0.8 \text{ m}$$

$$h = 0.2 \text{ m}$$

$$v_0 = 4.15 \text{ m/s}$$

$$\beta = 30^\circ$$

$$\dot{\theta}_m = 182.5 \text{ rad/s}$$